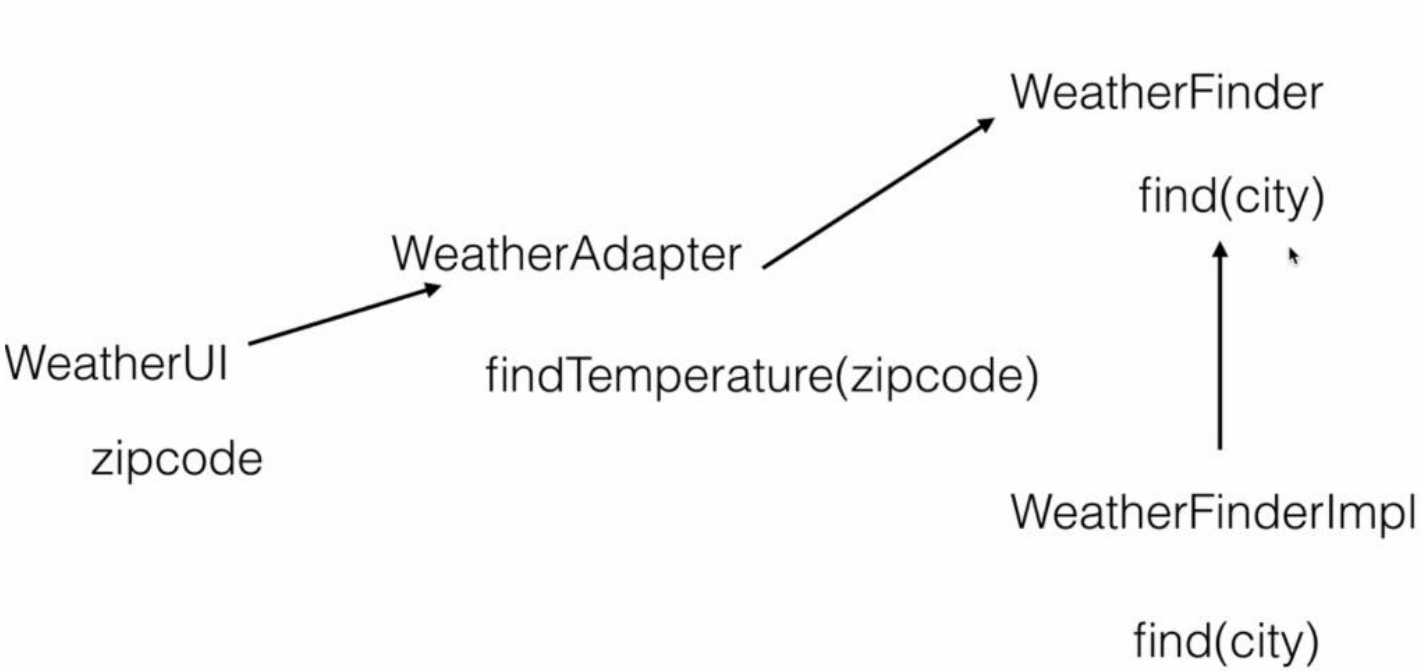
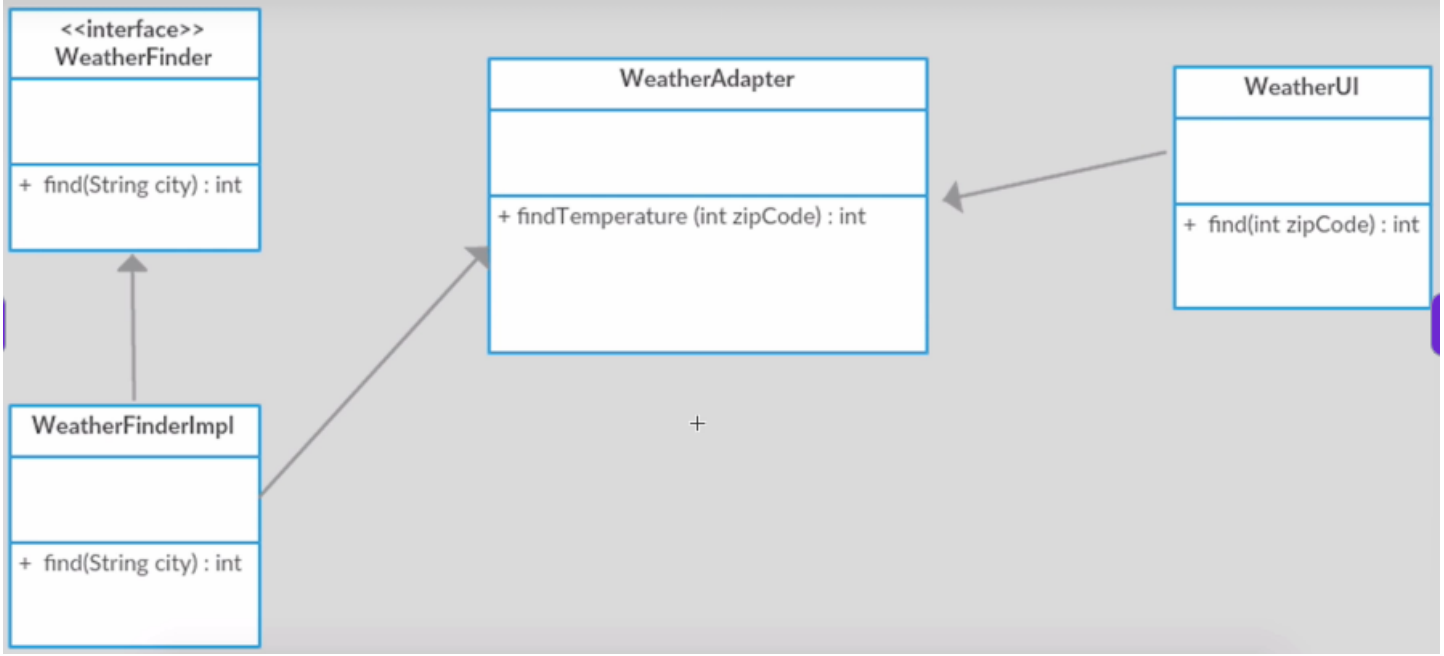
5-Adapter Pattern

- Adapter = “Translate one interface into another so classes can work together”

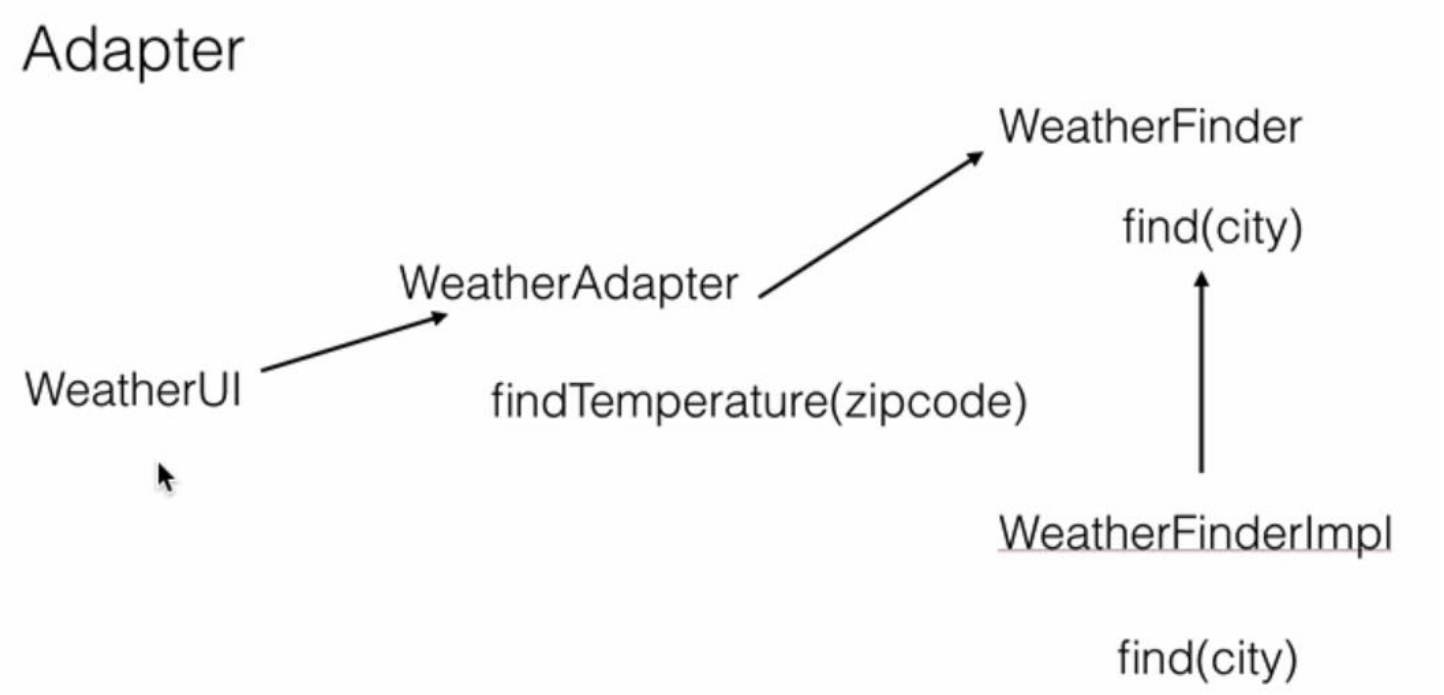
- When we have two applications communication: Similarly in the world of programming when we have two applications communicating with each other or two objects using each other and one object invokes the method off another object.



- សម្រាប់ adapter វាត្រូវតែមាន Two Class to communicate
- Weather Adapter វានៅកណ្តាលដើម្បីធ្វើការ ទំនាក់ទំនង រវាង Class: Weather UI ដែលត្រូវការ Zipcode to find the location and Class WeatherFinder វាត្រូវការ name of the city to fine the weather.



What we will design:



Purpose convert from String to Int: this is the adapter use for!

1- Create interface: WeatherFinder

udemy-course-design-pattern ▸ src/main/java ▸ com.vibol.corepattern.adapter ▸ WeatherFinder ▸

```java
package com.vibol.corepattern.adapter;

public interface WeatherFinder {

    int find(String city);

}
```

2- WeatherFinderImpl

udemy-course-design-pattern ▸ src/main/java ▸ com.vibol.corepattern.adapter ▸ WeatherFinderImpl ▸

```java
package com.vibol.corepattern.adapter;

public class WeatherFinderImpl implements WeatherFinder {

    // This is what we can from database
    @Override
    public int find(String city) {
        return 33;
    }
}
```

3- Adapter: WeatherAdapter

udemy-course-design-pattern ▸ src/main/java ▸ com.vibol.corepattern.adapter ▸ WeatherA

```java
package com.vibol.corepattern.adapter;

public class WeatherAdapter {

    public int findTemperature(int zipCode) {

        String city = null;

        if(zipCode == 12345) {

            city = "Phnom Penh";

        }

        WeatherFinder finder = new WeatherFinderImpl();

        int temperature = finder.find(city);

        return temperature;

    }

}
```

4- Weather UI

WeatherFinder.java | WeatherFinderImpl.java | WeatherAdapter.java | WeatherUI.java

udemy-course-design-pattern ▸ src/main/java ▸ com.vibol.corepattern.adapter ▸ WeatherUI ▸

```java
package com.vibol.corepattern.adapter;

public class WeatherUI {

    public void showTemperature(int zipCode) {

        WeatherAdapter adapter = new WeatherAdapter();
        System.out.println(adapter.findTemperature(zipCode));

    }

    public static void main(String[] args) {
        WeatherUI ui = new WeatherUI();
        ui.showTemperature(12345);
    }
}
```